Polyalphabetic Cipher + One Time Pad

requirements

- provided key has the same length of input plaintext
- key is shorter than input

+ Encryption

case 1: input and key has same length

given : A=0, B=1, C=2, ..., Z=25

formula: $C_i = (P_i + K_i) \bmod 26$

Plaintext: SMART

Key : SMILE

convert letters to numbers:

$$P = \begin{bmatrix} 18 \\ 12 \\ 0 \\ 17 \\ 19 \end{bmatrix} \qquad K = \begin{bmatrix} 18 \\ 12 \\ 8 \\ 11 \\ 4 \end{bmatrix}$$

Calculate each letter:

S + S = (19+18) mod 26 = 36 mod 26 = 10 = K

M + M = (12+12) mod 26 = 24 mod 26 = 24 = Y

A + I = (0+8) mod 26 = 8 mod 26 = 8 = I

R + L = (17+11) mod 26 = 28 mod 26 = 2 = C

T + E = (19+4) mod 26 = 23 mod 26 = 23 = X

Therefore Ciphertext = KYICX

case 2: key is shorter than input

plaintext: LAPEACE

key : SMILESM

calculate each letter:

L + S = (11+18) mod 26 = 29 mod 26 = 3 = D

A + M = (0+12) mod 26 = 12 mod 12 = 12 = M

P + I = (15+8) mod 26 = 23 mod 26 = 23 = X

E + L = (4+11) mod 26 = 15 mod 26 = 15 = P

A + E = (0+4) mod 26 = 4 mod 26 = 4 = E

C + S = (2+18) mod 26 = 20 mod 26 = 20 = U

E + M = (4+12) mod 26 = 16 mod 26 = 16 = Q

Therefore Plaintext = DMXPEUQ

+ Decryption

formula: $P_i = (C_i - K_i + 26) \bmod 26$

case 1:

ciphertext = KYICK

key = SMILE

calculate each letter:

K + S = (10 - 18 + 26) mod 26 = 18 mod 26 = 18 = S

Y + M = (24 - 12 + 26) mod 26 = 38 mod 26 = 12 = M

I + I = (8 - 8 + 26) mod 26 = 0 = A

C + L = (2 - 11 + 26) mod 26 = 17 mod 26 = 17 = R

K + E = (23 - 4 + 26) mod 26 = 45 mod 26 = 19 = T

Therefore Plaintext = SMART

case 2:

ciphertext: DMXPEQA

key : SMILESM

calculate each letter:

D - S = (3 - 18 + 26) mod 26 = 11 = L

M - M = (12 - 12 + 26) mod 26 = 0 = A

X - I = (23 - 8 + 26) mod 26 = 15 = P

P - L = (15 - 11 + 26) mod 26 = 4 = E

E - E = (4 - 4 + 26) mod 26 = 0 = A

Q - S = (20 - 18 + 26) mod 26 = 2 = C

Q - M = (16 - 12 + 26) mod 26 = 4 = E

Therefore Plaintext = LAPEACE